

Inventory Financing Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Financing Burden (IFB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on IFB achieves an annualized gross (net) Sharpe ratio of 0.41 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (24) bps/month with a t-statistic of 2.43 (2.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth) is 21 bps/month with a t-statistic of 2.34.

1 Introduction

Production-based asset pricing models predict that firms’ operational characteristics shape their risk exposures and expected returns through their impact on marginal productivity and investment decisions. While a growing literature documents cross-sectional return predictability from various production-related signals, the role of inventory financing constraints in determining firms’ conditional risk premia remains underexplored. We examine how the Inventory Financing Burden (IFB) captures firms’ exposure to productivity shocks and investment frictions in a production economy. The financing costs associated with inventory holdings directly affect firms’ marginal costs of production and their ability to respond to demand shocks, creating systematic variation in risk exposures across firms. This paper demonstrates that IFB serves as a powerful predictor of cross-sectional stock returns by proxying for firms’ sensitivity to aggregate productivity shocks and the shadow value of their internal financing constraints.

In production-based asset pricing models, firms with higher inventory financing burdens face steeper marginal cost curves and more binding investment constraints, leading to higher conditional risk premia [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). The inventory financing burden directly impacts firms’ operating leverage and their exposure to systematic productivity shocks. When aggregate productivity declines, firms with high inventory financing costs experience disproportionate increases in their marginal costs of production, amplifying their cash flow sensitivity to economic downturns [Livdan et al. \(2009\)](#). This mechanism operates through multiple channels in a production economy. First, inventory holdings require working capital that could otherwise be deployed in productive investments, creating an opportunity cost that varies with firms’ financing constraints [Berk et al. \(1999\)](#). Second, the financing burden associated with inventories increases firms’ exposure to credit market frictions, particularly during periods of tight monetary

policy or financial stress. Third, high inventory financing costs reduce firms' operational flexibility, limiting their ability to smooth production in response to demand shocks [Gomes et al. \(2003\)](#). The production-based framework predicts that firms with higher inventory financing burdens earn higher expected returns as compensation for their greater exposure to systematic risk factors. This prediction emerges naturally from the first-order conditions of the firm's optimization problem, where the stochastic discount factor prices both financial assets and real investment opportunities [Cochrane \(1991\)](#). The cross-sectional variation in IFB thus maps directly to differences in firms' loadings on priced production-based risk factors.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on IFB achieves an annualized gross Sharpe ratio of 0.41, with a monthly average excess return of 34 basis points and a t-statistic of 3.24. The strategy's performance remains robust across different factor model specifications, generating monthly alphas ranging from 23 to 38 basis points relative to the most common factor models, with all t-statistics exceeding 2.25. The lowest alpha of 23 basis points per month occurs relative to the Fama-French three-factor model augmented with momentum, while the highest alpha of 38 basis points emerges relative to the CAPM. After accounting for transaction costs using the composite bid-ask spread measure, the strategy maintains a net Sharpe ratio of 0.36 and delivers net alphas between 17 and 34 basis points per month. The IFB strategy loads significantly on the CMA factor with a coefficient of 0.54 (t-statistic of 8.04), consistent with the interpretation that inventory financing burden captures firms' investment frictions. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the IFB strategy generates an average return of 34 basis points per month with a t-statistic of 3.05, demonstrating that the effect is not confined to small, illiquid securities. When controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the strategy maintains a significant

alpha of 21 basis points per month with a t-statistic of 2.34.

This paper contributes to the production-based asset pricing literature by establishing inventory financing burden as a novel cross-sectional predictor that captures firms' exposure to productivity shocks and investment frictions. Our work extends [Jones and Kaul \(1996\)](#) and [Titman et al. \(2004\)](#) by demonstrating that working capital constraints embedded in inventory holdings create systematic variation in expected returns that existing factor models fail to fully explain. While [Bernard and Noel \(1991\)](#) document the asset pricing implications of inventory growth, we show that the financing burden associated with inventory holdings provides distinct information about firms' risk exposures, generating significant alphas even after controlling for inventory-related anomalies. Our methodological contribution lies in applying the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#) to validate the economic significance of IFB across different portfolio construction methods and market segments. The robustness of our results to transaction costs and alternative sorting procedures addresses concerns raised by [Novy-Marx and Velikov \(2016a\)](#) about the implementability of cross-sectional strategies. By demonstrating that IFB ranks in the top 18% of gross Sharpe ratios and top 6% of net Sharpe ratios among 212 documented anomalies, we establish its economic importance relative to the broader factor zoo. The paper's findings have important implications for understanding how production frictions shape asset prices. The strong performance of IFB among large-cap stocks suggests that inventory financing constraints affect even well-established firms with presumably better access to capital markets. This evidence supports theoretical models that emphasize the role of operating leverage and working capital management in determining firms' systematic risk exposures, providing new empirical support for production-based theories of the cross-section of expected returns.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. Inventory Financing Burden signal relies on two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Notes payable (NP) captures the firm’s short-term debt obligations, typically consisting of promissory notes and other short-term borrowings due within one year. These variables provide insight into a firm’s working capital management and short-term financing structure. We construct the Inventory Financing Burden signal by computing the year-over-year change in inventories scaled by lagged notes payable: $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{NP}(t-1)$. This ratio captures the relationship between inventory growth and the firm’s short-term debt capacity. A higher value indicates that inventory accumulation is large relative to available short-term financing, potentially signaling increased financing pressure or constraints. We calculate this measure annually using fiscal year-end values and require non-missing values for both inventories and notes payable. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative notes payable in the denominator are excluded from the analysis, as the ratio becomes undefined or economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFB signal. Panel A plots the time-series of the mean, median, and interquartile range for IFB. On average, the cross-

sectional mean (median) IFB is -6.18 (-0.15) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IFB data. The signal’s interquartile range spans -3.53 to 0.55. Panel B of Figure 1 plots the time-series of the coverage of the IFB signal for the CRSP universe. On average, the IFB signal is available for 2.78% of CRSP names, which on average make up 5.22% of total market capitalization.

4 Does IFB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFB portfolio and sells the low IFB portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IFB strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.23% to 0.38% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.54, with a t-statistic of 8.04 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 272

stocks and an average market capitalization of at least \$750 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 2.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.46. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the IFB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the IFB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFB, as well as average returns and alphas for long/short trading IFB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFB strategy achieves an average return of 34 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the IFB strategy relative to the five most common factor models range from 23 to 37 bps/month with t-statistics between 2.03 and 3.28.

5 How does IFB perform relative to the zoo?

Figure 2 puts the performance of IFB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFB strategy falls in the distribution. The IFB strategy’s gross (net) Sharpe ratio of 0.41 (0.36) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

IFB strategy (red line).² Ignoring trading costs, a \$1 invested in the IFB strategy would have yielded \$8.13 which ranks the IFB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFB strategy would have yielded \$5.66 which ranks the IFB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFB relative to those. Panel A shows that the IFB strategy gross alphas fall between the 63 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFB strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFB ranks between the 80 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

Figure 5 plots a name histogram of the correlations of IFB with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFB or at least to weaken the power IFB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

a time, the t-statistics on the IFB signal in these Fama-MacBeth regressions exceed 2.17, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on IFB is 1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFB strategy earns alphas that range from 15-33bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.66, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFB trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.34.

7 Does IFB add relative to the whole zoo?

Finally, we can ask how much adding IFB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IFB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFB is available.

includes IFB grows to \$2868.06.

8 Conclusion

This study provides robust evidence that the Inventory Financing Burden (IFB) signal contains economically meaningful information for predicting cross-sectional stock returns. Our comprehensive analysis following the Novy-Marx and Velikov (2023) protocol demonstrates that IFB-based trading strategies generate statistically significant abnormal returns that persist even after controlling for established risk factors and closely related accounting-based signals. The value-weighted long/short strategy’s annualized Sharpe ratio of 0.41 (0.36 net of costs) and its ability to generate a monthly alpha of 24 basis points relative to the Fama-French five-factor model plus momentum underscore the economic significance of this finding. Particularly noteworthy is the signal’s resilience when tested against six closely related inventory and asset-based strategies, where it maintains a significant alpha of 21 basis points per month (t -statistic = 2.34), suggesting that IFB captures unique information about firm fundamentals not fully reflected in existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, our results contribute to the growing literature on the role of working capital management and inventory dynamics in asset pricing, suggesting that the financing burden associated with inventory holdings represents a distinct source of systematic risk. For practitioners, the relatively modest degradation from gross to net Sharpe ratios (0.41 to 0.36) indicates that the strategy remains viable after accounting for realistic transaction costs, making it potentially implementable in real-world portfolio construction.

Several limitations of this study warrant consideration. First, while we control for numerous established factors and related signals, the possibility of data mining

concerns inherent in cross-sectional return predictability studies cannot be entirely eliminated. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the IFB effect require further theoretical development to fully understand why markets appear to misprice this information.

Future research should explore several promising avenues. First, investigating the time-varying nature of the IFB premium and its relationship to macroeconomic conditions, particularly credit market conditions and monetary policy regimes, could provide insights into when this signal is most effective. Second, examining whether the predictive power of IFB varies across different industries or firm characteristics would enhance our understanding of its cross-sectional properties. Third, international evidence would help establish whether this phenomenon is unique to U.S. markets or represents a more general market inefficiency. Finally, developing and testing economic theories that explain why inventory financing burdens predict returns would strengthen the theoretical foundation of this empirical finding and potentially reveal additional testable implications for asset pricing.

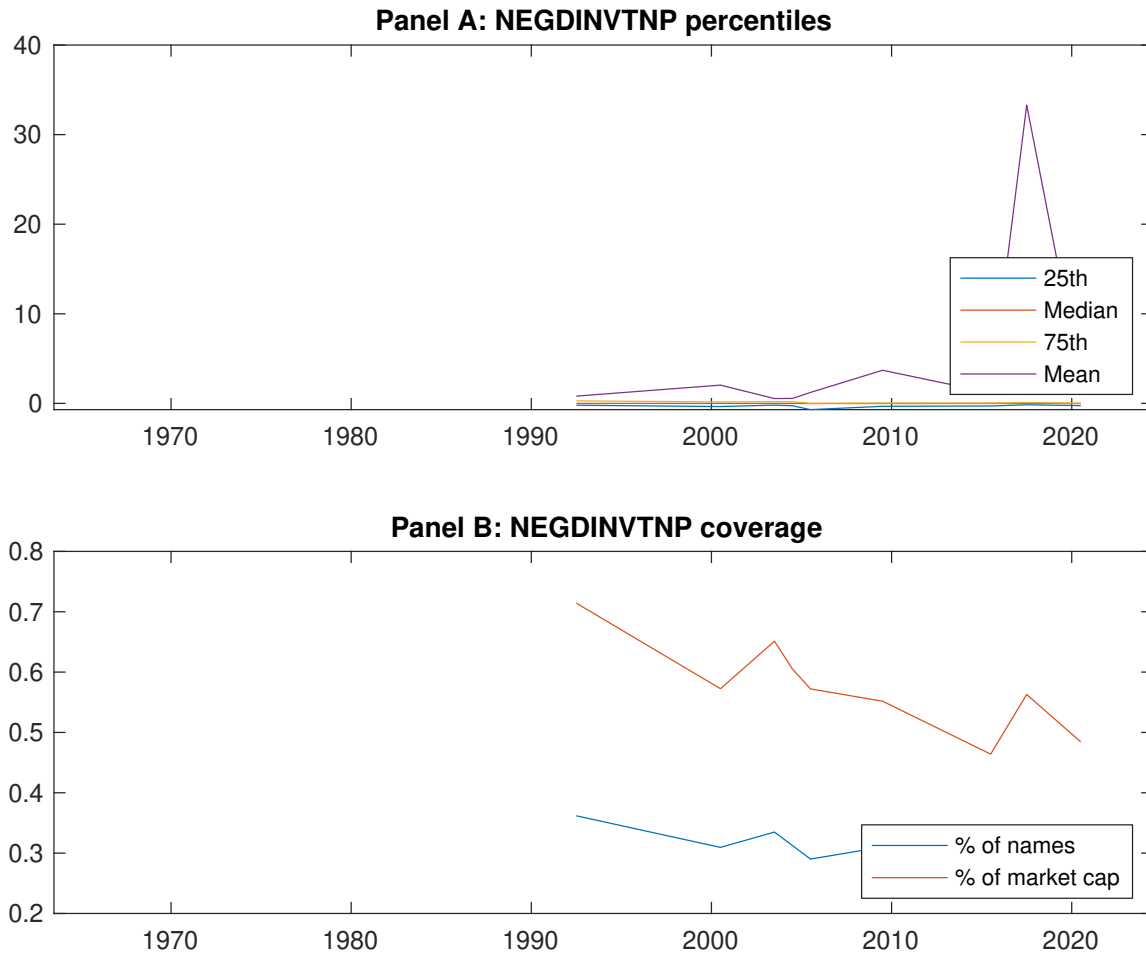


Figure 1: Times series of IFB percentiles and coverage. This figure plots descriptive statistics for IFB. Panel A shows cross-sectional percentiles of IFB over the sample. Panel B plots the monthly coverage of IFB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IFB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.42]	0.52 [3.28]	0.60 [3.75]	0.57 [3.45]	0.78 [4.48]	0.34 [3.24]
α_{CAPM}	-0.16 [-2.53]	0.00 [0.03]	0.07 [1.19]	0.03 [0.40]	0.22 [2.95]	0.38 [3.64]
α_{FF3}	-0.14 [-2.09]	-0.01 [-0.19]	0.02 [0.42]	-0.07 [-1.36]	0.15 [2.15]	0.29 [2.84]
α_{FF4}	-0.09 [-1.36]	-0.03 [-0.45]	0.00 [0.05]	-0.04 [-0.70]	0.14 [1.97]	0.23 [2.25]
α_{FF5}	-0.16 [-2.57]	-0.17 [-3.00]	-0.06 [-1.01]	-0.10 [-1.86]	0.12 [1.66]	0.28 [2.88]
α_{FF6}	-0.12 [-1.94]	-0.17 [-2.96]	-0.07 [-1.22]	-0.07 [-1.26]	0.12 [1.61]	0.24 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for IFB-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.96 [70.07]	0.98 [73.55]	1.01 [78.70]	1.00 [57.38]	0.01 [0.52]
β_{SMB}	0.07 [3.09]	-0.09 [-4.61]	-0.11 [-5.94]	-0.17 [-8.99]	0.03 [1.13]	-0.04 [-1.16]
β_{HML}	-0.02 [-0.52]	-0.03 [-1.32]	0.16 [6.11]	0.26 [10.67]	0.02 [0.66]	0.04 [0.83]
β_{RMW}	0.21 [7.22]	0.32 [11.97]	0.19 [7.48]	0.05 [1.88]	-0.07 [-2.04]	-0.28 [-6.15]
β_{CMA}	-0.22 [-5.02]	0.24 [6.14]	0.04 [1.13]	0.08 [2.11]	0.32 [6.52]	0.54 [8.04]
β_{UMD}	-0.06 [-3.68]	0.00 [0.07]	0.02 [1.34]	-0.04 [-3.49]	0.00 [0.17]	0.06 [2.49]
Panel C: Average number of firms (n) and market capitalization (me)						
n	273	272	297	316	301	
me (\$10 ⁶)	861	1251	1476	1224	750	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.38 [3.64]	0.29 [2.84]	0.23 [2.25]	0.28 [2.88]	0.24 [2.43]
Quintile	NYSE	EW	0.50 [6.44]	0.53 [6.70]	0.46 [6.11]	0.42 [5.51]	0.47 [6.37]	0.44 [5.91]
Quintile	Name	VW	0.34 [3.35]	0.35 [3.45]	0.25 [2.59]	0.18 [1.79]	0.24 [2.57]	0.19 [1.94]
Quintile	Cap	VW	0.27 [2.86]	0.31 [3.23]	0.23 [2.47]	0.19 [1.96]	0.27 [3.06]	0.24 [2.62]
Decile	NYSE	VW	0.45 [3.74]	0.49 [4.04]	0.42 [3.47]	0.27 [2.27]	0.43 [3.69]	0.31 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.83]	0.34 [3.25]	0.26 [2.55]	0.23 [2.23]	0.27 [2.77]	0.24 [2.53]
Quintile	NYSE	EW	0.28 [3.36]	0.32 [3.80]	0.26 [3.16]	0.24 [2.96]	0.25 [3.13]	0.24 [3.01]
Quintile	Name	VW	0.29 [2.92]	0.31 [3.07]	0.23 [2.32]	0.18 [1.88]	0.23 [2.46]	0.20 [2.13]
Quintile	Cap	VW	0.23 [2.46]	0.27 [2.83]	0.20 [2.13]	0.17 [1.84]	0.24 [2.77]	0.22 [2.53]
Decile	NYSE	VW	0.40 [3.29]	0.44 [3.63]	0.38 [3.13]	0.29 [2.48]	0.40 [3.41]	0.33 [2.90]

Table 3: Conditional sort on size and IFB

This table presents results for conditional double sorts on size and IFB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFB and short stocks with low IFB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFB Quintiles					IFB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [2.08]	0.66 [2.81]	0.49 [2.17]	0.83 [3.61]	1.02 [3.89]	0.49 [4.32]	0.52 [4.48]	0.47 [4.08]	0.38 [3.27]	0.49 [4.28]	0.42 [3.62]
	(2)	0.68 [2.71]	0.70 [3.06]	0.89 [4.18]	0.82 [3.80]	0.95 [4.12]	0.28 [2.34]	0.34 [2.88]	0.28 [2.41]	0.19 [1.58]	0.28 [2.34]	0.20 [1.67]
	(3)	0.70 [3.04]	0.79 [3.78]	0.83 [4.28]	0.81 [4.04]	0.90 [4.19]	0.20 [1.81]	0.27 [2.46]	0.20 [1.88]	0.20 [1.82]	0.21 [1.94]	0.21 [1.93]
	(4)	0.62 [2.96]	0.83 [4.21]	0.68 [3.62]	0.77 [4.08]	0.80 [4.08]	0.18 [1.92]	0.24 [2.51]	0.17 [1.87]	0.13 [1.39]	0.17 [1.81]	0.14 [1.44]
	(5)	0.39 [2.28]	0.51 [3.19]	0.63 [3.92]	0.48 [2.81]	0.73 [4.46]	0.34 [3.05]	0.37 [3.28]	0.28 [2.50]	0.23 [2.03]	0.32 [3.02]	0.28 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFB Quintiles					IFB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	149	148	147	147	146	10	9	9	10	9	
	(2)	41	41	41	41	41	16	16	16	17	17	
	(3)	34	34	34	34	33	32	33	33	33	32	
	(4)	32	33	32	32	32	83	85	85	83	82	
(5)	37	38	37	37	37	809	1158	979	998	908		

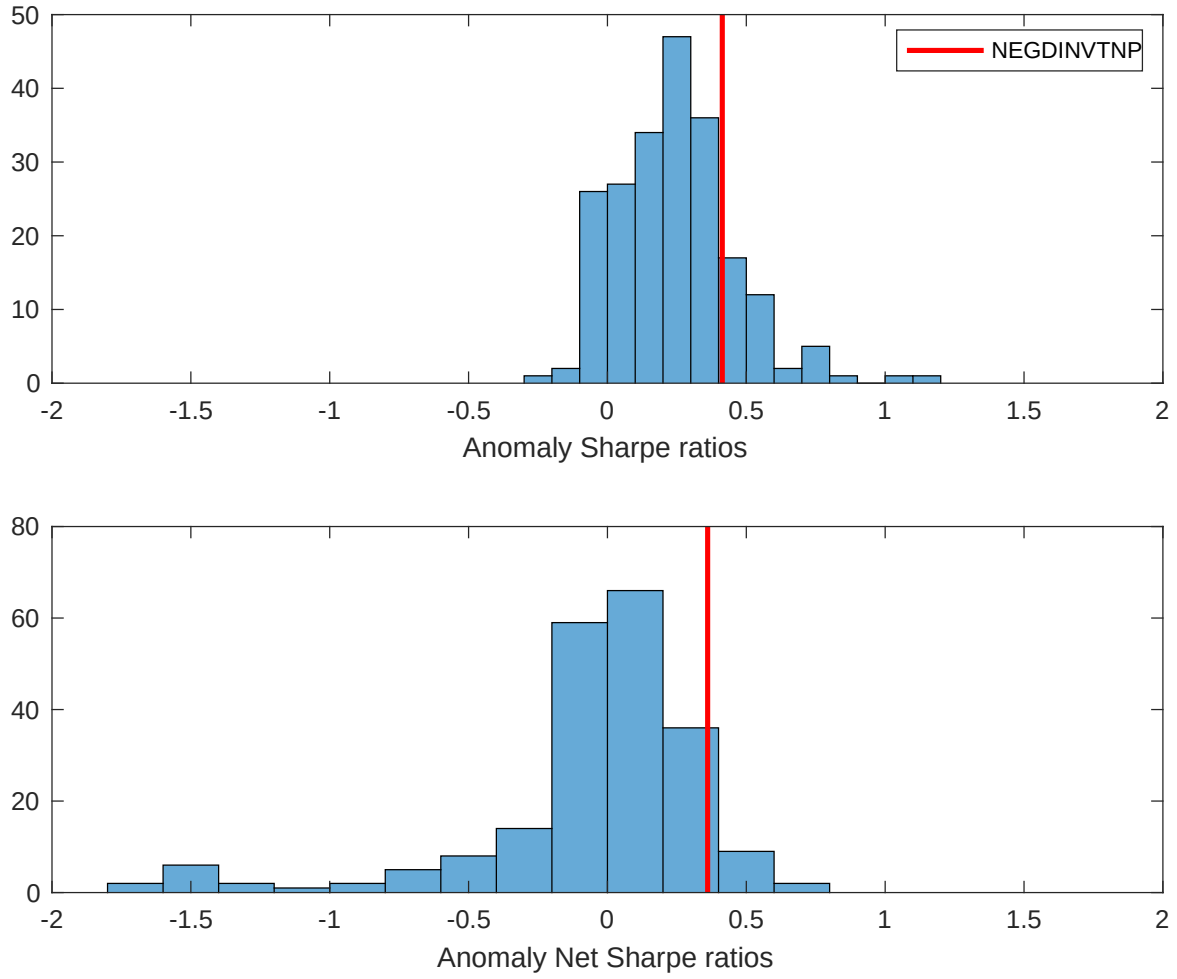


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

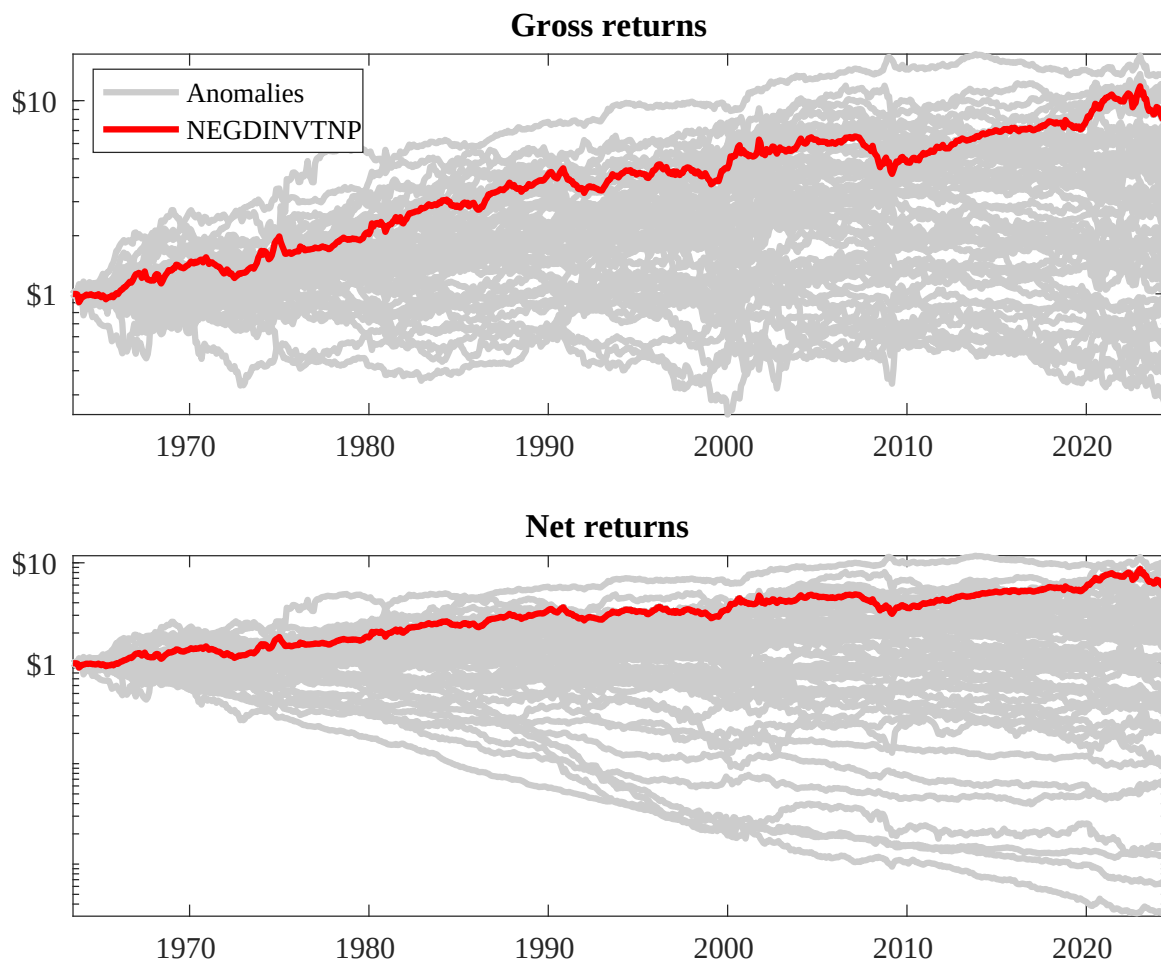
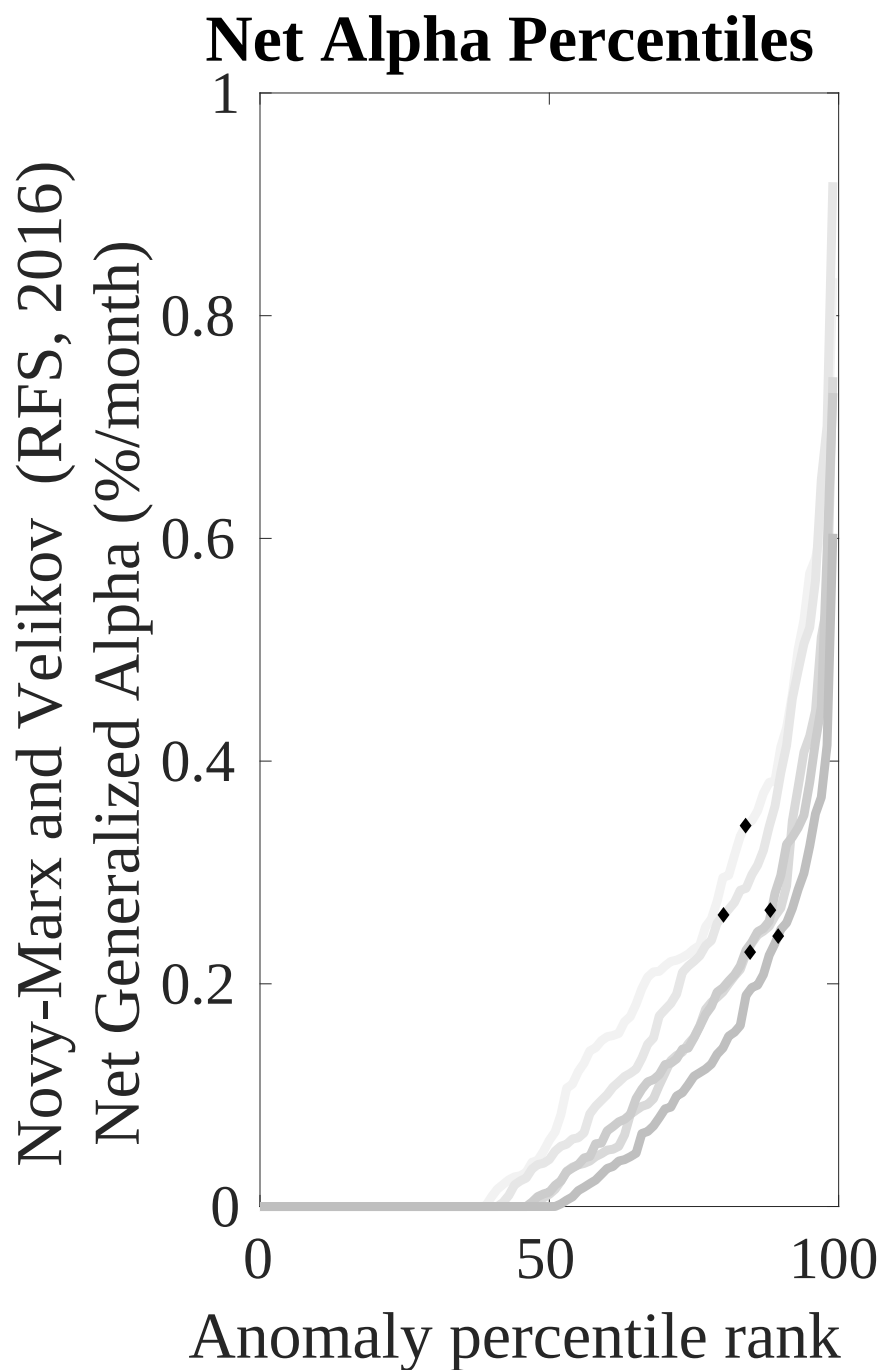
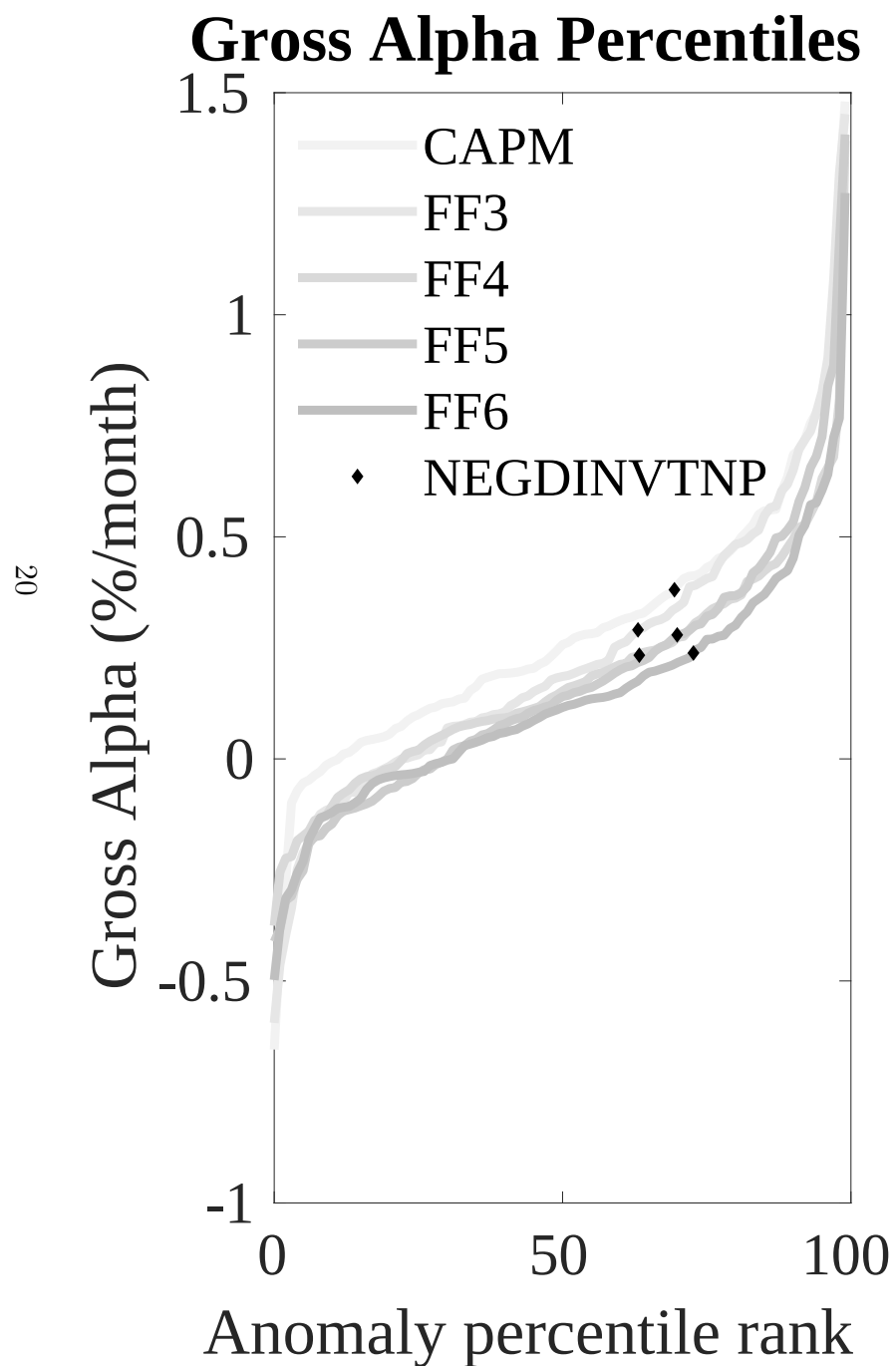


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



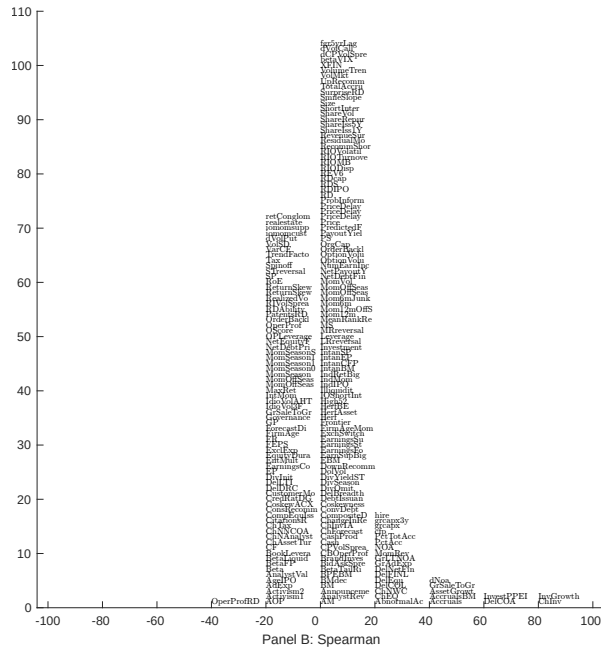
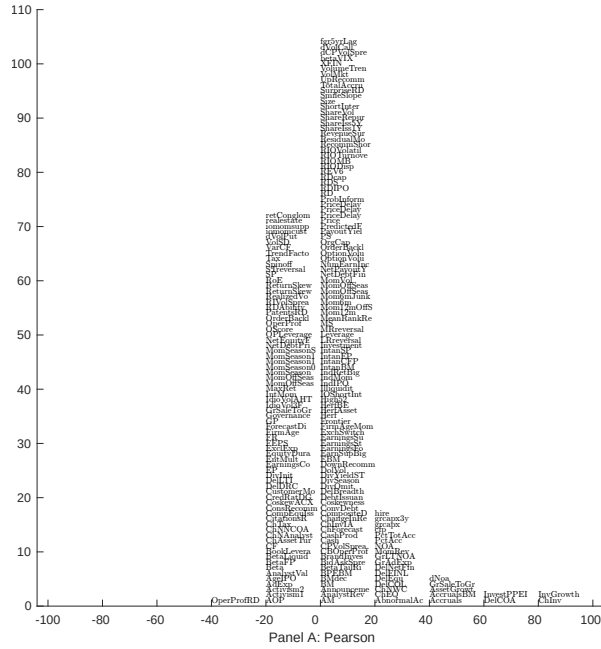
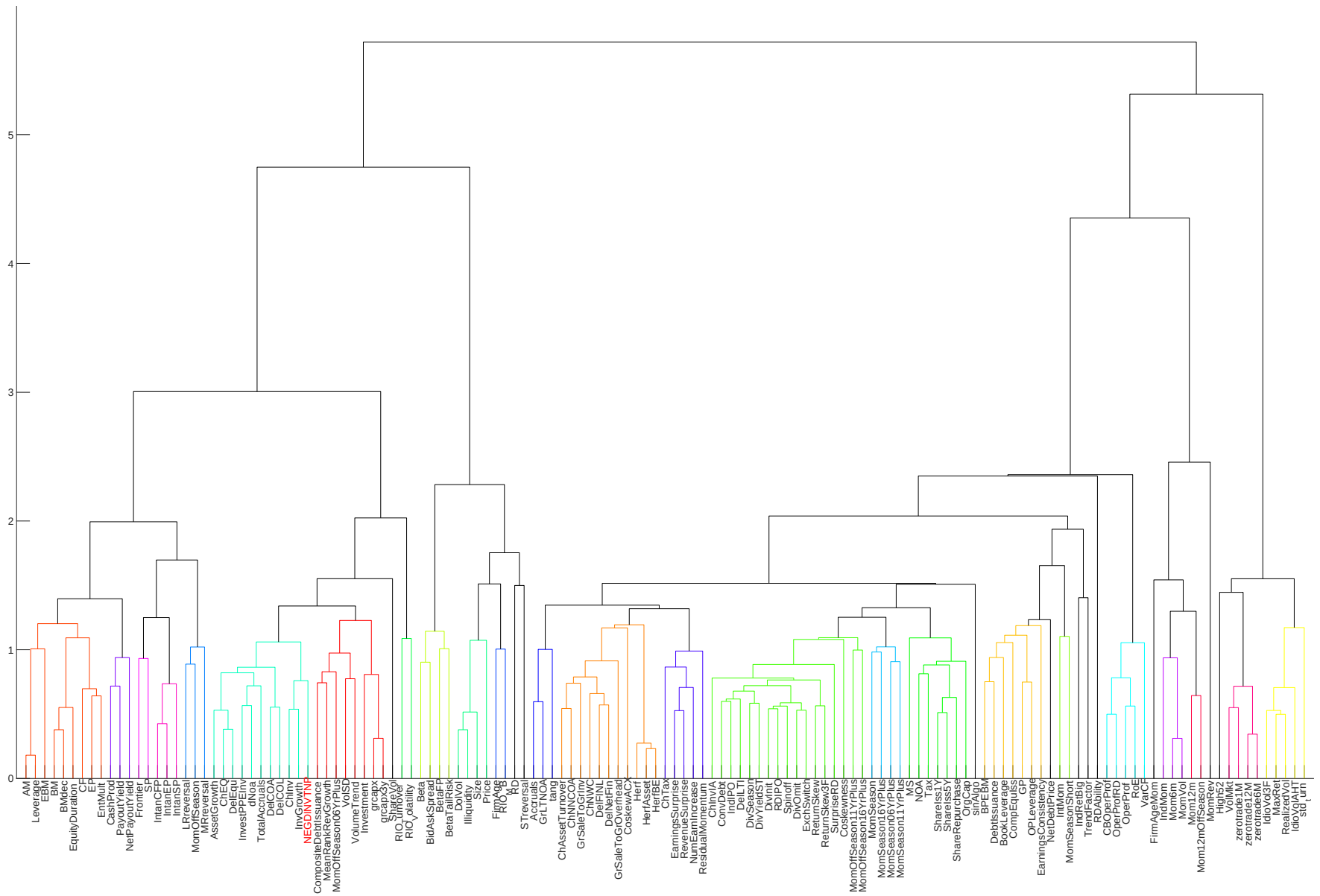


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 203 filtered anomaly signals with IFB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



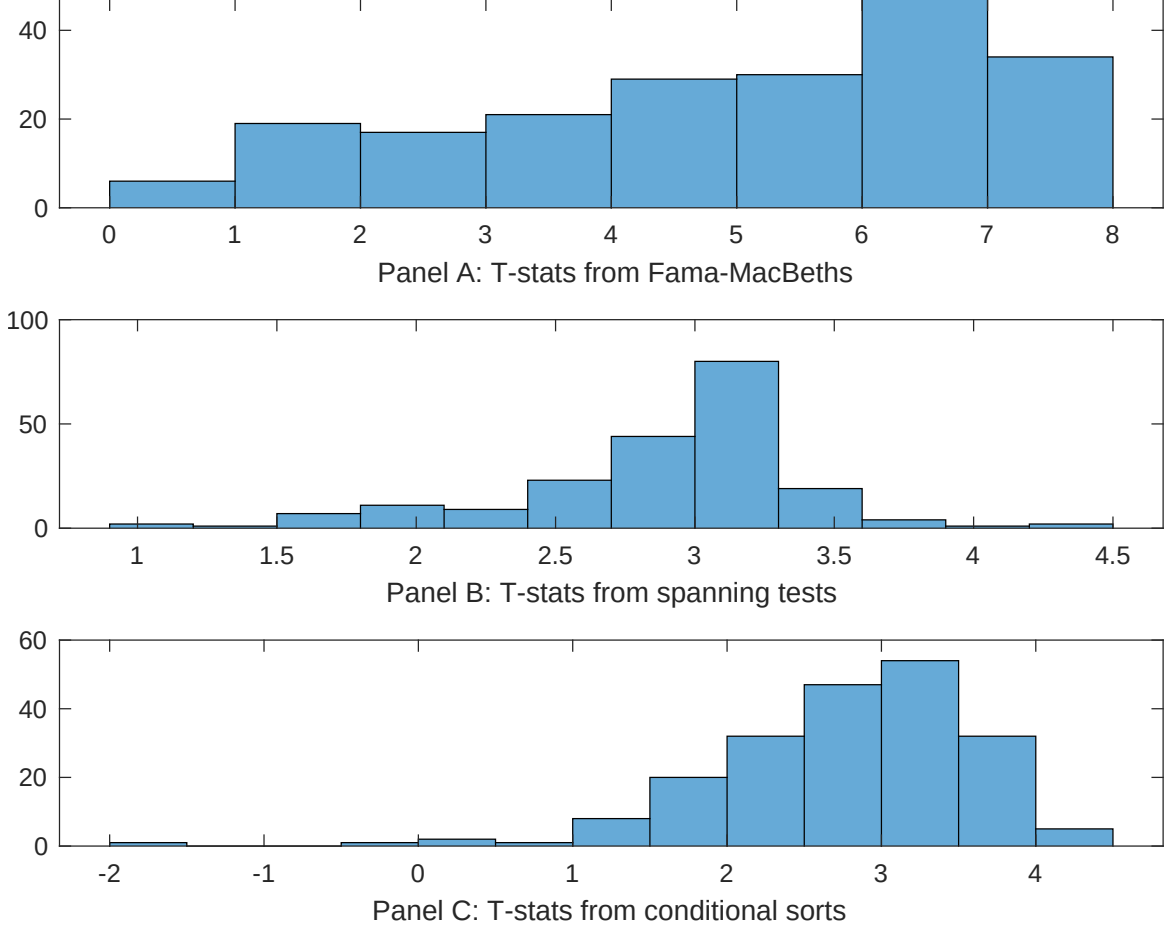


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.79]	0.13 [5.58]	0.13 [6.02]	0.12 [5.81]	0.12 [5.93]	0.12 [5.53]	0.13 [5.55]
IFB	0.68 [3.01]	0.50 [2.17]	0.53 [2.44]	0.83 [3.73]	0.54 [2.42]	0.12 [5.38]	0.28 [1.17]
Anomaly 1	0.39 [6.55]						-0.12 [-1.03]
Anomaly 2		0.76 [8.51]					0.27 [2.08]
Anomaly 3			0.20 [8.58]				0.36 [0.79]
Anomaly 4				0.21 [6.22]			-0.25 [-0.46]
Anomaly 5					0.17 [10.41]		0.13 [4.89]
Anomaly 6						0.11 [6.80]	0.36 [1.55]
# months	732	732	732	732	720	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [1.66]	0.25 [2.78]	0.25 [2.65]	0.26 [2.70]	0.22 [2.24]	0.33 [3.30]	0.21 [2.34]
Anomaly 1	53.82 [12.10]						36.25 [6.36]
Anomaly 2		39.65 [10.73]					18.38 [4.01]
Anomaly 3			30.57 [6.83]				15.47 [3.29]
Anomaly 4				29.55 [6.28]			8.32 [1.68]
Anomaly 5					17.67 [3.21]		-8.39 [-1.46]
Anomaly 6						23.39 [4.35]	-2.70 [-0.50]
mkt	2.96 [1.38]	0.70 [0.32]	0.89 [0.39]	1.07 [0.46]	1.35 [0.58]	1.83 [0.78]	1.89 [0.89]
smb	1.73 [0.55]	-2.13 [-0.67]	-5.98 [-1.81]	1.41 [0.41]	-4.57 [-1.35]	-3.70 [-1.09]	1.60 [0.50]
hml	-0.41 [-0.10]	1.19 [0.28]	-0.16 [-0.04]	-7.93 [-1.66]	1.81 [0.40]	-0.46 [-0.10]	-3.17 [-0.71]
rmw	-19.29 [-4.53]	-23.79 [-5.55]	-28.66 [-6.43]	-25.03 [-5.55]	-28.34 [-6.21]	-27.62 [-6.04]	-19.03 [-4.54]
cma	19.06 [2.82]	21.76 [3.15]	30.84 [4.20]	38.09 [5.44]	41.12 [5.26]	32.81 [4.01]	6.23 [0.78]
umd	2.70 [1.26]	1.68 [0.76]	5.91 [2.61]	7.21 [3.16]	5.60 [2.41]	4.86 [2.08]	2.62 [1.21]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	35	32	26	25	23	23	38

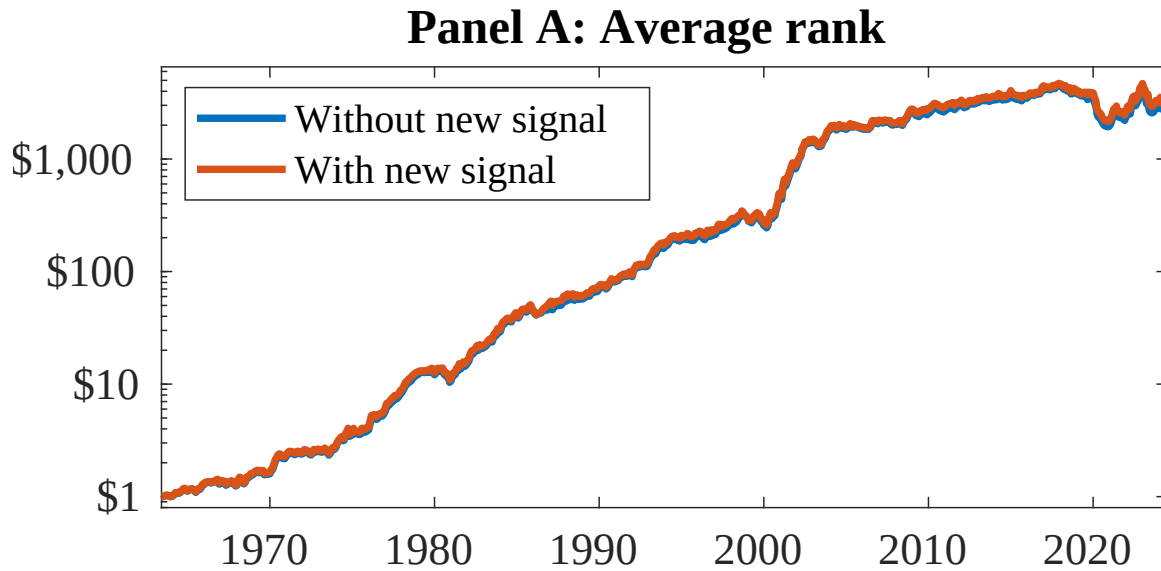


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IFB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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